Hossein Razizadeh

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Experience

2022–2024 **Senior Software Engineer**, OVOU, Vancouver, Canada.

- **Back Office System Design and Implementation**: Led design and implementation of a back-office system that streamlined fulfillment workflows and made processes **4x faster**.
- **Backend Development**: Built and optimized **Django REST APIs** serving **100K+ users**, refactored legacy code, and led data migrations to improve performance, scalability, and system reliability.
- ▷ *Technologies:* Python, Django REST Framework, PostgreSQL, Pytest, Docker, Kubernetes

2016–2021 **Senior Software Engineer**, Cafe Bazaar, Tehran, Iran.

- **Computer Vision Service Development**: Collaborated with data scientists to develop ML-powered computer vision services, including a watermark detection system processing **over 5 million images daily**, and optimized backend performance **by 30%**.
- **POI Extraction Pipeline Development**: Achieved a **98.5% accuracy** rate in extracting Points of Interest (POI) from images captured by street-imaging cars, utilizing a combination of automated models and human annotations.
- **Street Imaging Project Management**: Led the Street Imaging project, building multi-camera vehicle systems capturing 30 fps imagery, and processed **200M+ street images** to enrich Balad map's POI database.
- **Interview Process Improvement**: Improved the technical hiring process by refining candidate profiles, updating interview problems to reflect real-world challenges, and introducing questions targeting analytical and system design skills.
- **Mentorship & Team Support**: Mentored and facilitated new team members, promoting skill growth, collaboration, and knowledge transfer.
- ▷ *Technologies:* Python, Django, gRPC, Kubernetes, Docker, PostgreSQL, C++, S3, Ceph, Linux, Parallel Processing

2014–2016 **Senior Software Engineer**, Sabz Biomedicals, Tehran, Iran.

Pioneered development of high-performance karyotyping and 2D gel analysis software, leading a small team and architecting core systems to improve lab diagnostics and genetic analysis accuracy.
Technologies: C++, Qt, OpenCV, Windows, Image Processing

2007–2008 **Software Engineer**, Pooya Fanavaran Kosar, Tehran, Iran.

Engineered a C++-based in-car speed camera system, boosting detection accuracy and reliability; earned 1st-Place Award in Applied Research at the Young Khwarizmi Festival.
Technologies: C++, OpenCV, Socket Programming, Pipes, Inter-Process Communication, Windows

2004–2006 **Software Engineer, RoboCup Team Aria**, Amirkabir University of Technology, Tehran, Iran.

Built real-time multi-agent systems in **C++** for the 3D Soccer Simulation League, developing autonomous strategy and coordination logic on Linux. Contributed to the team's success among **30+ international teams** in RoboCup competitions.
Technologies: C++, Linux, Multi-Agent Systems, Real-Time Systems, AI Strategy

2005 **Software Engineering Internship**, Rabi Kosar, Tehran, Iran.

Developed specialized code in C and Embedded-C to facilitate lossless high-speed signal transmission from DSP boards to PCs, enhancing data integrity and processing efficiency.
Technologies: C, Embedded-C, DSP Framework, Real-Time Signal Processing, Semaphores, Parallel Processing

2003–2016 **Computer Group Manager and Lecturer**, Allameh Helli High School, Tehran, Iran.

Managed a team of 10 computer science teachers and taught courses in programming, algorithms and data structures.
Technologies: C++, Pascal, Algorithm Design, Data Structures, Discrete Mathematics, Graph Theory

Awards and Honors

2005 **World Championship Award**, *RoboCup 2005*, Osaka, Japan.
1st place among 32 international teams in 3D Soccer Simulation league

2007, 2006, 2005 **5th, 6th, 6th Team Rank**, *ACM ICPC West Asia Region Regional Contest*, Tehran, Iran.

Education

MSc in Computer Science, *The University of Tehran*, Tehran, Iran.
Thesis: Development and Implementation of an Image to Speech System to be used in Persian language.

BSc in Computer Engineering, *Amirkabir University of Technology, Tehran Polytechnic*, Tehran, Iran.
Thesis: Evaluation of reinforcement learning algorithms on control problems with delayed reward signals.

Skills

Computer Skills
Programming Languages: **C++, C, Python**
Technologies: **Linux, Docker, Kubernetes, gRPC, STL, OpenCV, Git, Django, Unit Testing(Pytest), GIS, Windows, Microsoft Visual Studio, Scrum**

Analytical Skills
Data Structure, Design and Analysis of Algorithms, Parallel Processing, Design Patterns, System Design, TDD, OOP, Artificial Neural Networks, Genetic Algorithm, Image Processing